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Orthogonal Parametric Non-negative Matrix Tri-Factorization with α -Divergence for Co-clustering

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- Our algorithm works by multiplicative update rules and it is convergence.
- Adding two penalties for controlling the orthogonality of row and column clusters.
- Unifying a class of algorithms for co-clustering based on α -divergence.
- [NMTFcoclust](#) GitHub repository provides all materials and additional visualizations.

Orthogonal Parametric Non-negative Matrix Tri-Factorization with α -Divergence for Co-clustering

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Abstract

Co-clustering algorithms can seek homogeneous sub-matrices into a dyadic data matrix, such as a document-word matrix. Algorithms for co-clustering can be expressed as a non-negative matrix tri-factorization problem such that $\mathbf{X} \approx \mathbf{FSG}^\top$, which is associated with the non-negativity conditions on all matrices and the orthogonality of $\mathbf{F}$ (row-coefficient) and $\mathbf{G}$ (column-coefficient) matrices. Most algorithms are based on Euclidean distance and Kullback-Leibler divergence without parameters to control orthogonality. We propose to apply the orthogonality of parameters by adding two penalty terms based on the α -divergence objective function. Orthogonal parametric non-negative matrix tri-factorization uses orthogonal parameters for row and column space, separately. Finally, we compare the proposed algorithms with other algorithms on six real text datasets.

Keywords: Co-clustering, Dyadic data, α -divergence, Non-negative matrix tri-factorization, Orthogonality, Text mining.

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1. Introduction

Dyadic data matrices, such as document-word counts, movie-viewer ratings, and product-customer purchases matrices, refer to the duality between rows and columns that frequently arise in various essential applications—for example, collaborative filtering (Shan and Banerjee, 2010), text mining (Hotho et al., 2005), and gene expression data analysis (Hofmann et al., 1999). A fundamental problem in dyadic data analysis is discovering the hidden sub-matrices (Long et al., 2005; Del Buono and Pio, 2015). Seeking sub-matrices in a dyadic data matrix is an old idea called co-clustering.

Traditional one-side clustering cannot seek a relation between sub-sets of rows (documents) and columns (words), but simultaneously clustering on rows and columns is useful for discovering sub-matrices into a dyadic data matrix (Hartigan, 1972; Govaert and Nadif, 2013). Matrix factorization methods have been widely used in dyadic data analysis (Ding et al., 2006; Yoo and Choi, 2010).

Non-negative Matrix Factorization (NMF) (Lee and Seung, 1999; Seung and Lee, 2001; Pauca et al., 2006) and Non-negative Matrix Tri-Factorization (NMTF) are both important non-negative low rank approximation techniques. NMF is a framework for clustering, and co-clustering can be formulated by NMTF (Wang and Zhang, 2012; Li and Ding, 2018).

Non-negative Block Value Decomposition (NBVD) (Long et al., 2005) introduced a three non-negative factor decomposition for $\mathbf{X}$ as a document-word matrix by the multiplication of $\mathbf{F}$, $\mathbf{S}$, and $\mathbf{G}$ as the matrix of row coefficients, block value, and column coefficients, respectively. Orthogonal Non-negative Matrix 3-Factorization (ONM3F) is a three factorization of a non-negative $\mathbf{X}$ with orthogonality constraints ($\mathbf{F}^\top \mathbf{F} = \mathbf{I}_g$ and $\mathbf{G}^\top \mathbf{G} = \mathbf{I}_s$)

(Ding et al., 2006). Orthogonal Non-negative Matrix Tri-Factorization (ONMTF) (Yoo and Choi, 2010) developed a multiplicative update rule algorithm with orthogonality constraints. The co-clustering of document-word matrices by NBVD, ONM3F, and ONMTF was demonstrated to be effective. Double NMTF (DNMTF) and Orthogonal Double NMTF (ODNMTF) were proposed in (Labiod and Nadif, 2011), with the idea of finding two factors matrices rather than three factors matrices, which summarized matrix $\mathbf{S} = \mathbf{F}^\top \mathbf{X} \mathbf{G}$. Penalized Non-negative Matrix Tri-Factorization (PNMTF) introduced three penalty terms to replace the three orthogonality constraints (Wang and Huang, 2017).

These methods find factor matrices $\mathbf{F}$ and $\mathbf{G}$ based on Euclidean distance or Kullback–Leibler divergence as an objective function. The orthogonality of $\mathbf{F}$ and $\mathbf{G}$ guarantees the uniqueness of the matrices (Ding et al., 2006) and also applies independent constraints among labels of rows (columns) from a statistical perspective.

In recent years, various measures of dissimilarity, such as Bregman divergences (Banerjee et al., 2007), Csiszár’s f -divergences, and Amari’s α -divergences, have been examined under NMF (Cichocki et al., 2008, 2006a,b; Sra and Dhillon, 2005). Multiplicative update rules algorithms were proposed in (Cichocki et al., 2006c), based on α -divergence (Amari, 2012) as a particular case of Csiszár’s f -divergence. NMF multiplicative algorithm based on α -divergence was proposed in (Cichocki et al., 2008) for image denoising and EEG classification. Suppose that $\mathbf{X}$ is a document-word matrix, which factor matrices obtained using (1) are represented in Figure 1.

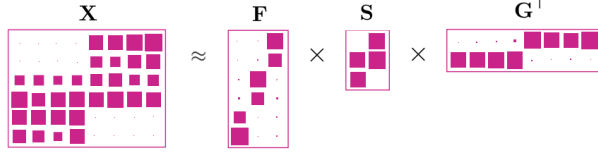


Figure 1: Similar to (Yoo and Choi, 2010), a graphical illustration of document-word co-clustering $\mathbf{X} \in \mathbb{R}_+^{6 \times 8}$. An element in the matrix with a bigger square has a larger value. NMTF identifies three factors $\mathbf{F}$, $\mathbf{S}$, and $\mathbf{G}$ for three rows (document) clusters and two columns (word) clusters. $\mathbf{S}$ is a matrix that shows the connections between document-clusters and word-clusters. Each column of $\mathbf{S}$ indicates which document-clusters contribute to the word-cluster. The second and third document-cluster are related to the first word-cluster, also the first and second document-cluster are related to the second word-cluster.

The main contributions of this paper are summarized as follows:

- We proposed a co-clustering algorithm Orthogonal Parametric Non-negative Matrix Tri-Factorization (OPNMTF) by Adding two penalty terms for controlling the orthogonality of row and column clusters based on α -divergence.
- We use the α -divergence as a measure of divergence between the observation matrix and the approximation matrix. This unification permits more flexibility in determining divergence measures by changing the value of α .
- Experiments on six real text datasets demonstrate the effectiveness of the proposed model compared to the state-of-the-art co-clustering methods.

In this paper, we present NMTF based on α -divergence and the probabilistic interpretation of NMTF in Section 2. We introduce a novel OPNMTF in Section 3, with convergence proof and multiplicative update rules by matrix-wise form. Also, we demonstrate that our algorithm is derivable

using the Karush-Kuhn-Tucker conditions. In Section 4, we implement competitor algorithms listed in Table 1 on six real text datasets. All notations are listed in Appendix A.

Table 1: The comparison between algorithms according to the optimization problem and multiplicative update rules.

Algorithm	Optimization problem	Multiplicative update rules
OPNMTF (Proposed 3)	$\arg \min_{\mathbf{F} \geq 0, \mathbf{S} \geq 0, \mathbf{G} \geq 0} D_{\alpha}(\mathbf{X} \ \mathbf{FSG}^{\top}) + \lambda D_{\beta}(\mathbf{L}_y \ \mathbf{F}^{\top} \mathbf{F}) + \mu D_{\alpha}(\mathbf{L}_x \ \mathbf{G}^{\top} \mathbf{G})$	$\mathbf{F} \odot \left[\frac{(\mathbf{X} \odot \mathbf{FSG}^{\top})^{\alpha} \mathbf{GS}^{\top} + 2\lambda \mathbf{F}(\mathbf{L}_y \odot \mathbf{F}^{\top} \mathbf{F})^{\alpha}}{\mathbf{E}_{mm} \mathbf{GS}^{\top} + 2\lambda \mathbf{F} \mathbf{L}_y} \right]^{\frac{1}{\alpha}},$ $\mathbf{G} \odot \left[\frac{((\mathbf{X} \odot \mathbf{FSG}^{\top})^{\top})^{\alpha} \mathbf{FS} + 2\mu \mathbf{G}(\mathbf{L}_x \odot \mathbf{G}^{\top} \mathbf{G})^{\alpha}}{\mathbf{E}_{mm} \mathbf{FS} + 2\mu \mathbf{G} \mathbf{L}_x} \right]^{\frac{1}{\alpha}},$ $\mathbf{S} \odot \left[\frac{\mathbf{F}^{\top} (\mathbf{X} \odot \mathbf{FSG}^{\top})^{\alpha} \mathbf{G}}{\mathbf{F}^{\top} \mathbf{E}_{mm} \mathbf{G}} \right]^{\frac{1}{\alpha}}.$
PNMTF (Wang and Huang, 2017)	$\arg \min_{\mathbf{F} \geq 0, \mathbf{S} \geq 0, \mathbf{G} \geq 0} \frac{1}{2} \ \mathbf{X} - \mathbf{FSG}^{\top}\ ^2 + \frac{\tau}{2} \text{Tr}(\mathbf{F} \Psi_y \mathbf{F}^{\top}) + \frac{\eta}{2} \text{Tr}(\mathbf{G} \Psi_x \mathbf{G}^{\top}) + \frac{\gamma}{2} \text{Tr}(\mathbf{S}^{\top} \mathbf{S})$	$\mathbf{F} \odot \left[\frac{\mathbf{XGS}^{\top}}{\mathbf{FSG}^{\top} \mathbf{GS}^{\top} + \tau \mathbf{F} \Psi_y} \right]^{\frac{1}{2}}, \mathbf{G} \odot \left[\frac{\mathbf{X}^{\top} \mathbf{FS}}{\mathbf{GS}^{\top} \mathbf{F}^{\top} \mathbf{FS} + \eta \mathbf{G} \Psi_x} \right]^{\frac{1}{2}},$ $\mathbf{S} \odot \left[\frac{\mathbf{F}^{\top} \mathbf{XG}}{\mathbf{F}^{\top} \mathbf{FSG}^{\top} \mathbf{G} + \gamma \mathbf{S}} \right]^{\frac{1}{2}}.$
DNMF (Labiod and Nadif, 2011)	$\arg \min_{\mathbf{F}^{\top} \mathbf{F} = \mathbf{I}_p, \mathbf{G}^{\top} \mathbf{G} = \mathbf{I}_s} \ \mathbf{X} - \mathbf{FF}^{\top} \mathbf{XGG}^{\top}\ ^2$	$\mathbf{F} \odot \left[\frac{\mathbf{XGG}^{\top} \mathbf{X}^{\top} \mathbf{F}}{\mathbf{FF}^{\top} \mathbf{XGG}^{\top} \mathbf{X}^{\top} \mathbf{F}} \right]^{\frac{1}{2}},$ $\mathbf{G} \odot \left[\frac{\mathbf{X}^{\top} \mathbf{FF}^{\top} \mathbf{XG}}{\mathbf{GG}^{\top} \mathbf{X}^{\top} \mathbf{FF}^{\top} \mathbf{XG}} \right]^{\frac{1}{2}}.$
ODNMF (Labiod and Nadif, 2011)	$\arg \min_{\mathbf{F} \geq 0, \mathbf{S} \geq 0, \mathbf{G} \geq 0} \ \mathbf{X} - \mathbf{FF}^{\top} \mathbf{XGG}^{\top}\ ^2 + \text{Tr}(\mathbf{A} \mathbf{F}^{\top} \mathbf{F} - \mathbf{I}_p) + \text{Tr}(\mathbf{I}_p \mathbf{G}^{\top} \mathbf{G} - \mathbf{I}_s)$	$\mathbf{F} \odot \left[\frac{2\mathbf{XGG}^{\top} \mathbf{X}^{\top} \mathbf{F}}{\mathbf{FF}^{\top} \mathbf{XGG}^{\top} \mathbf{GG}^{\top} \mathbf{X}^{\top} \mathbf{F} + \mathbf{XGG}^{\top} \mathbf{GG}^{\top} \mathbf{X}^{\top} \mathbf{FF}^{\top} \mathbf{F}} \right]^{\frac{1}{2}},$ $\mathbf{G} \odot \left[\frac{2\mathbf{X}^{\top} \mathbf{FF}^{\top} \mathbf{XG}}{\mathbf{GG}^{\top} \mathbf{FF}^{\top} \mathbf{XX}^{\top} \mathbf{FF}^{\top} \mathbf{G} + \mathbf{FF}^{\top} \mathbf{XX}^{\top} \mathbf{FF}^{\top} \mathbf{GG}^{\top} \mathbf{G}} \right]^{\frac{1}{2}},$ $\mathbf{F} \odot \left[\frac{\mathbf{XGS}^{\top}}{\mathbf{FSG}^{\top} \mathbf{X}^{\top} \mathbf{F}} \right]^{\frac{1}{2}}, \mathbf{G} \odot \left[\frac{\mathbf{X}^{\top} \mathbf{FS}}{\mathbf{GS}^{\top} \mathbf{F}^{\top} \mathbf{XG}} \right]^{\frac{1}{2}},$ $\mathbf{S} \odot \left[\frac{\mathbf{F}^{\top} \mathbf{XG}}{\mathbf{F}^{\top} \mathbf{FSG}^{\top} \mathbf{G}} \right]^{\frac{1}{2}}.$
ONMTF (Yoo and Choi, 2010)	$\arg \min_{\mathbf{F} \geq 0, \mathbf{S} \geq 0, \mathbf{G} \geq 0, \mathbf{F}^{\top} \mathbf{F} = \mathbf{I}_p, \mathbf{G}^{\top} \mathbf{G} = \mathbf{I}_s} \frac{1}{2} \ \mathbf{X} - \mathbf{FSG}^{\top}\ ^2$	$\mathbf{F} \odot \left[\frac{(\mathbf{XGS}^{\top})^{\frac{1}{2}}}{(\mathbf{FF}^{\top} \mathbf{XGS}^{\top})^{\frac{1}{2}}} \right]^{\frac{1}{2}}, \mathbf{G} \odot \left[\frac{(\mathbf{X}^{\top} \mathbf{FS})^{\frac{1}{2}}}{(\mathbf{GG}^{\top} \mathbf{X}^{\top} \mathbf{FS})^{\frac{1}{2}}} \right]^{\frac{1}{2}},$ $\mathbf{S} \odot \left[\frac{\mathbf{F}^{\top} \mathbf{XG}}{\mathbf{F}^{\top} \mathbf{FSG}^{\top} \mathbf{G}} \right]^{\frac{1}{2}}.$
ONMF3 (Ding et al., 2006)	$\arg \min_{\mathbf{F} \geq 0, \mathbf{S} \geq 0, \mathbf{G} \geq 0, \mathbf{F}^{\top} \mathbf{F} = \mathbf{I}_p, \mathbf{G}^{\top} \mathbf{G} = \mathbf{I}_s} \ \mathbf{X} - \mathbf{FSG}^{\top}\ ^2 + \text{Tr}(\mathbf{A}(\mathbf{F}^{\top} \mathbf{F} - \mathbf{I}_p)) + \text{Tr}(\mathbf{I}_p(\mathbf{G}^{\top} \mathbf{G} - \mathbf{I}_s))$	$\mathbf{F} \odot \left[\frac{(\mathbf{F}^{\top} \mathbf{XG})^{\frac{1}{2}}}{(\mathbf{F}^{\top} \mathbf{FSG}^{\top} \mathbf{G})^{\frac{1}{2}}} \right]^{\frac{1}{2}},$ $\mathbf{F} \odot \left[\frac{\mathbf{XGS}^{\top}}{\mathbf{FSG}^{\top} \mathbf{GS}^{\top}} \right]^{\frac{1}{2}}, \mathbf{G} \odot \left[\frac{\mathbf{X}^{\top} \mathbf{FS}}{\mathbf{GS}^{\top} \mathbf{F}^{\top} \mathbf{FS}} \right]^{\frac{1}{2}},$ $\mathbf{S} \odot \left[\frac{\mathbf{F}^{\top} \mathbf{XG}}{\mathbf{F}^{\top} \mathbf{FSG}^{\top} \mathbf{G}} \right]^{\frac{1}{2}}.$
NBVD (Long et al., 2005)	$\arg \min_{\mathbf{F} \geq 0, \mathbf{S} \geq 0, \mathbf{G} \geq 0} \ \mathbf{X} - \mathbf{FSG}^{\top}\ ^2$	$\mathbf{F} \odot \left[\frac{\mathbf{XGS}^{\top}}{\mathbf{FSG}^{\top} \mathbf{GS}^{\top}} \right]^{\frac{1}{2}}, \mathbf{G} \odot \left[\frac{\mathbf{X}^{\top} \mathbf{FS}}{\mathbf{GS}^{\top} \mathbf{F}^{\top} \mathbf{FS}} \right]^{\frac{1}{2}},$ $\mathbf{S} \odot \left[\frac{\mathbf{F}^{\top} \mathbf{XG}}{\mathbf{F}^{\top} \mathbf{FSG}^{\top} \mathbf{G}} \right]^{\frac{1}{2}}.$

* (Wang and Huang, 2017) τ , η , and γ are parameters orthogonality, also $\Psi_y = \mathbf{E}_{yy} - \mathbf{I}_y$ and $\Psi_x = \mathbf{E}_{xx} - \mathbf{I}_x$. (Labiod and Nadif, 2011) $\mathbf{A}$ and $\mathbf{I}$ are matrices containing Lagrangian multipliers. (Long et al., 2005) proposed two versions when $\mathbf{S} = \mathbf{F}^{\top} \mathbf{XG}$, also DNMF is called Double k-means.

2. Non-negative Matrix Tri-Factorization

Given observation matrix $\mathbf{X} = (X_{ij}) \in \mathbb{R}_+^{n \times m}$ and approximation matrix $\mathbf{FSG}^{\top} = ([\mathbf{FSG}^{\top}]_{ij}) \in \mathbb{R}_+^{n \times m}$. NMTF aims to find three matrices $\mathbf{F} = (F_{ik}) \in \mathbb{R}_+^{n \times g}$, $\mathbf{S} = (S_{kh}) \in \mathbb{R}_+^{g \times s}$, and $\mathbf{G} = (G_{jh}) \in \mathbb{R}_+^{m \times s}$ with non-negative elements. The low-rank approximation of $\mathbf{X}$ by

$$\mathbf{X} \approx \mathbf{FSG}^{\top}, \quad (1)$$

where n , m , $g \leq n$, and $s \leq m$ are the numbers of rows, columns, row clusters, and column clusters, respectively. Also, $\mathbf{F}$, $\mathbf{G}$, and $\mathbf{S}$ play roles in membership row clustered, column clustered, and summarization matrix, respectively. This triple decomposition enables a suitable framework for simultaneous clustering on $\mathbf{X}$.

Definition 1. The α -divergence between $\mathbf{X}$ and $\mathbf{FSG}^\top$ is defined as follows (Cichocki et al., 2008):

$$\begin{aligned} D_\alpha(\mathbf{X} \parallel \mathbf{FSG}^\top) &= \frac{1}{\alpha(1-\alpha)} \sum_{i,j} \left(\alpha X_{ij} + (1-\alpha)[\mathbf{FSG}^\top]_{ij} - X_{ij}^\alpha [\mathbf{FSG}^\top]_{ij}^{1-\alpha} \right) \\ &= \sum_{i,j} X_{ij} f\left(\frac{[\mathbf{FSG}^\top]_{ij}}{X_{ij}}\right), \end{aligned} \quad (2)$$

where $f(\cdot)$ is a convex function for positive values α as follows

$$f(y) = \frac{1}{\alpha(1-\alpha)} \left(\alpha + (1-\alpha)y - y^{1-\alpha} \right). \quad (3)$$

The α -divergence (2) includes Kullback-Leibler (KL), Hellinger, and Pearson- χ^2 divergences. The divergence criteria for special values of α are listed in Table 2, and their corresponding functions (3) are plotted in Figure 2.

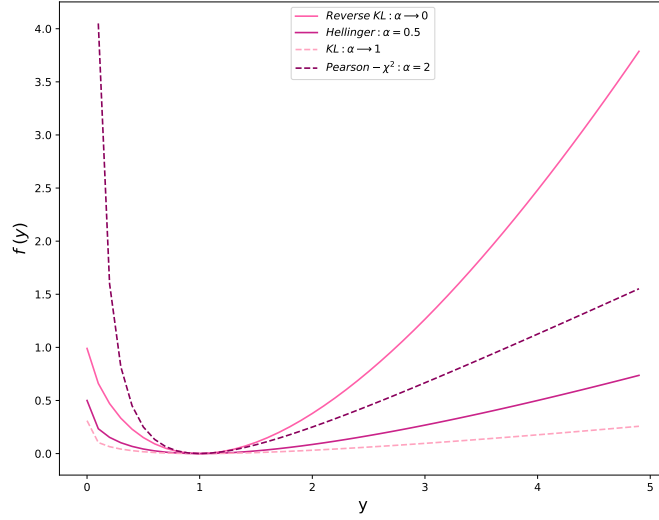


Figure 2: The convex function $f(y)$ in (3) for various values.

Table 2: Types of α -divergence for $\mathbf{X}$ as data matrix and $\mathbf{FSG}^\top$ as approximation matrix.

Divergence	α	$D_\alpha(\mathbf{X} \parallel \mathbf{FSG}^\top)$
KL	1^*	$\sum_{i,j} \left(X_{ij} \log \left(\frac{X_{ij}}{[\mathbf{FSG}^\top]_{ij}} \right) - X_{ij} + [\mathbf{FSG}^\top]_{ij} \right)$
Hellinger	0.5	$2 \sum_{i,j} \left(\sqrt{X_{ij}} - \sqrt{[\mathbf{FSG}^\top]_{ij}} \right)^2$
Pearson- χ^2	2	$\frac{1}{2} \sum_{i,j} \frac{(X_{ij} - [\mathbf{FSG}^\top]_{ij})^2}{[\mathbf{FSG}^\top]_{ij}}$

* $\lim_{\alpha \rightarrow 1} D_\alpha(\mathbf{X} \parallel \mathbf{FSG}^\top)$ equal to KL, and $\lim_{\alpha \rightarrow 0} D_\alpha(\mathbf{X} \parallel \mathbf{FSG}^\top)$ equal to reverse KL.

Multiplicative update rules are efficient algorithms for solving the underlying optimization problem

$$\arg \min_{\mathbf{F} \geq 0, \mathbf{S} \geq 0, \mathbf{G} \geq 0} D_\alpha(\mathbf{X} \parallel \mathbf{FSG}^\top). \quad (4)$$

It is crucial to introduce an auxiliary function to analyze convergence and

to derive algorithms.

Definition 2. $A(\mathbf{F}, \mathbf{F}^{(t)})$ is called an auxiliary function for $D_\alpha(\mathbf{X} \parallel \mathbf{F} \mathbf{S} \mathbf{G}^\top)$ as a function of $\mathbf{F}$ if conditions (5) and (6) are satisfied:

$$A(\mathbf{F}, \mathbf{F}) = D_\alpha(\mathbf{X} \parallel \mathbf{F} \mathbf{S} \mathbf{G}^\top), \quad (5)$$

$$A(\mathbf{F}, \mathbf{F}^{(t)}) \geq D_\alpha(\mathbf{X} \parallel \mathbf{F} \mathbf{S} \mathbf{G}^\top), \quad (6)$$

where $\mathbf{F}^{(t)}$ denotes the iteration t that resulted in the updating of $\mathbf{F}$.

Lemma 2.1. *If $A(\mathbf{F}^{(t+1)}, \mathbf{F}^{(t)})$ is an auxiliary function for $D_\alpha(\mathbf{X} \parallel \mathbf{F}^{(t+1)} \mathbf{S} \mathbf{G}^\top)$, then $D_\alpha(\mathbf{X} \parallel \mathbf{F}^{(t+1)} \mathbf{S} \mathbf{G}^\top)$ is non-increasing function of $\mathbf{F}^{(t+1)}$ with respect to the updating rule:*

$$\mathbf{F}^{(t+1)} = \arg \min_{\mathbf{F}} A(\mathbf{F}, \mathbf{F}^{(t)}). \quad (7)$$

Proof. See [Appendix B](#). □

2.1. Probabilistic interpretation of NMTF

A comprehensive discussion of the probabilistic interpretation of NMTF and NMF can be found in ([Gaussier and Goutte, 2005](#); [Yoo and Choi, 2010](#)). Assuming the k -th document cluster and the h -th word cluster are dc_k and wc_h respectively. $\mathbb{P}(d_i, w_j)$ as the joint probability document i and word j can be computed as follows:

$$\begin{aligned} \mathbb{P}(d_i, w_j) &= \sum_k \sum_h \mathbb{P}(d_i, w_j \mid dc_k, wc_h) \mathbb{P}(dc_k, wc_h) \\ &= \sum_k \sum_h \mathbb{P}(d_i \mid dc_k) \mathbb{P}(dc_k, wc_h) \mathbb{P}(w_j \mid wc_h). \end{aligned}$$

Suppose that $\mathbf{X}$ has been normalized to meet $\sum_{i,j} X_{ij} = 1$. $\mathbf{D}_{\mathbf{F}} = \text{diag}(\mathbf{1}_n^\top \mathbf{F})$ and $\mathbf{D}_{\mathbf{G}} = \text{diag}(\mathbf{1}_m^\top \mathbf{G})$ are diagonal matrices as sum-to-one normalization

and $\mathbf{1}_n^\top = (1, 1, \dots, 1) \in \mathbb{R}^n$. Therefore, decomposition (1) is rewritten as follows:

$$\mathbf{X} = \left(\mathbf{F} \mathbf{D}_{\mathbf{F}}^{-1} \right) \left(\mathbf{D}_{\mathbf{F}} \mathbf{S} \mathbf{D}_{\mathbf{G}} \right) \left(\mathbf{G} \mathbf{D}_{\mathbf{G}}^{-1} \right)^\top,$$

where $[\mathbf{F} \mathbf{D}_{\mathbf{F}}^{-1}]_{ik} = \mathbb{P}(d_i | dc_k)$, $[\mathbf{D}_{\mathbf{F}} \mathbf{S} \mathbf{D}_{\mathbf{G}}]_{kh} = \mathbb{P}(dc_k, wc_h)$, $[\mathbf{G} \mathbf{D}_{\mathbf{G}}^{-1}]_{jh} = \mathbb{P}(w_j | wc_h)$.

The posterior probability of document-cluster given documents is computed as

$$\begin{aligned} \mathbb{P}(dc_k | d_i) &\propto \mathbb{P}(d_i | dc_k) \mathbb{P}(dc_k) = \mathbb{P}(d_i | dc_k) \sum_h \mathbb{P}(dc_k, wc_h) \\ &= [\mathbf{F} \mathbf{D}_{\mathbf{F}}^{-1} \text{diag}(\mathbf{D}_{\mathbf{F}} \mathbf{S} \mathbf{D}_{\mathbf{G}} \mathbf{1}_m)]_{ik} \\ &= [\mathbf{F} \text{diag}(\mathbf{S} \mathbf{D}_{\mathbf{G}} \mathbf{1}_m)]_{ik}. \end{aligned}$$

Document i belongs to document-cluster k^* if

$$k^* = \arg \max_k [\mathbf{F} \text{diag}(\mathbf{S} \mathbf{D}_{\mathbf{G}} \mathbf{1}_m)]_{ik}.$$

For word clustering, we calculate the posterior probability of word-cluster given words as

$$\begin{aligned} \mathbb{P}(wc_h | w_j) &\propto \mathbb{P}(w_j | wc_h) \mathbb{P}(wc_h) = \mathbb{P}(w_j | wc_h) \sum_k \mathbb{P}(dc_k, wc_h) \\ &= [\text{diag}(\mathbf{1}_n^\top \mathbf{D}_{\mathbf{F}} \mathbf{S} \mathbf{D}_{\mathbf{G}}) (\mathbf{D}_{\mathbf{G}}^{-1} \mathbf{G}^\top)]_{hj} \\ &= [\text{diag}(\mathbf{1}_n^\top \mathbf{D}_{\mathbf{F}} \mathbf{S}) \mathbf{G}^\top]_{hj}. \end{aligned}$$

Word j belongs to word-cluster h^* if

$$h^* = \arg \max_h [\mathbf{G} \text{diag}(\mathbf{1}_n^\top \mathbf{D}_{\mathbf{F}} \mathbf{S})]_{jh}.$$

By probabilistic interpretation, Algorithm 1 will represent the updating rules obtained in the next section for clustering and co-clustering.

3. Orthogonal Parametric Non-negative Matrix Tri-Factorization

Orthogonality $\mathbf{F}$ and $\mathbf{G}$ can be achieved without considering restrictions ($\mathbf{F}^\top \mathbf{F} = \mathbf{I}_g$, $\mathbf{G}^\top \mathbf{G} = \mathbf{I}_s$). This goal is applied in Orthogonal Parametric Non-negative Matrix Tri-Factorization (OPNMTF). The objective function in OPNMTF has a two-term penalty based on α -divergence. The first term is for the row space with the control parameter λ . The second term is for the column space with the control parameter μ . We introduce our objective function:

$$\xi_{\text{OPNMTF}} = D_\alpha(\mathbf{X} \parallel \mathbf{FSG}^\top) + \lambda D_\alpha(\mathbf{I}_g \parallel \mathbf{F}^\top \mathbf{F}) + \mu D_\alpha(\mathbf{I}_s \parallel \mathbf{G}^\top \mathbf{G}) \quad (8)$$

Lemma 3.1. *If $\mathbf{G}$ and $\mathbf{S}$ are fixed, then the auxiliary function for ξ_{OPNMTF} is*

$$A(\mathbf{F}, \mathbf{F}^{(t)}) = \sum_{i,j,k,h} X_{ij} Q_{ijkh}^{\mathbf{F}^{(t)}} f\left(\frac{F_{ik} S_{kh} G_{hj}^\top}{X_{ij} Q_{ijkh}^{\mathbf{F}^{(t)}}}\right) + \lambda \sum_{i,k} Q_{ik}^{\mathbf{F}^{(t)}} f\left(\frac{F_{ki}^\top F_{ik}}{Q_{ik}^{\mathbf{F}^{(t)}}}\right) + \text{Const}_{\mathbf{G}}, \quad (9)$$

where $Q_{ijkh}^{\mathbf{F}^{(t)}} = \frac{F_{ik}^{(t)} S_{kh} G_{hj}^\top}{\sum_{k',h'} F_{ik'}^{(t)} S_{k'h'} G_{h'j}^\top}$, $Q_{ik}^{\mathbf{F}^{(t)}} = \frac{F_{ki}^{(t)\top} F_{ik}^{(t)}}{\sum_{i'} F_{ki'}^{(t)\top} F_{i'k}^{(t)}}$, and $\text{Const}_{\mathbf{G}}$ is a constant value and depends on $\mathbf{G}$, and $f(\cdot)$ is defined in (3).

Proof. See [Appendix C](#). □

Lemma 3.2. *If $\mathbf{F}$ and $\mathbf{S}$ are fixed, then the auxiliary function for ξ_{OPNMTF}*

is

$$A(\mathbf{G}, \mathbf{G}^{(t)}) = \sum_{i,j,k,h} X_{ij} Q_{ijkh}^{\mathbf{G}^{(t)}} f \left(\frac{F_{ik} S_{kh} G_{hj}^\top}{X_{ij} Q_{ijkh}^{\mathbf{G}^{(t)}}} \right) + \mu \sum_{j,h} Q_{jh}^{\mathbf{G}^{(t)}} f \left(\frac{G_{hj}^\top G_{jh}}{Q_{jh}^{\mathbf{G}^{(t)}}} \right) + \text{Const}_{\mathbf{F}}, \quad (10)$$

where $Q_{ijkh}^{\mathbf{G}^{(t)}} = \frac{F_{ik} S_{kh} G_{hj}^\top}{\sum_{k',h'} F_{ik'} S_{k'h'} G_{h'j}^\top}$, $Q_{jh}^{\mathbf{G}^{(t)}} = \frac{G_{hj}^\top G_{jh}}{\sum_{j'} G_{hj'}^\top G_{j'h}}$, and $\text{Const}_{\mathbf{F}}$ is a constant value and depends on $\mathbf{F}$, and $f(\cdot)$ is defined in (3).

Proof. It is the same as Lemma 3.1. $\square$

Lemma 3.3. If $\mathbf{F}$ and $\mathbf{G}$ are fixed, then the auxiliary function for ξ_{OPNMTF} is

$$A(\mathbf{S}, \mathbf{S}^{(t)}) = \sum_{i,j,k,h} X_{ij} Q_{ijkh}^{\mathbf{S}^{(t)}} f \left(\frac{F_{ik} S_{kh} G_{hj}^\top}{X_{ij} Q_{ijkh}^{\mathbf{S}^{(t)}}} \right) + \text{Const}_{\mathbf{F}, \mathbf{G}}, \quad (11)$$

where $Q_{ijkh}^{\mathbf{S}^{(t)}} = \frac{F_{ik} S_{kh} G_{hj}^\top}{\sum_{k',h'} F_{ik'} S_{k'h'} G_{h'j}^\top}$, $\text{Const}_{\mathbf{F}, \mathbf{G}}$ is a constant value and depends on $\mathbf{F}$ and $\mathbf{G}$, and $f(\cdot)$ is defined in (3).

Proof. It is the same as Lemma 3.1. $\square$

Theorem 3.4. The multiplicative update rule for ξ_{OPNMTF} (8) is given as follows:

$$\mathbf{F}^{(t+1)} \leftarrow \mathbf{F}^{(t)} \odot \left[\left(\frac{(\mathbf{X} \odot \mathbf{F} \mathbf{S} \mathbf{G}^\top)^\alpha \mathbf{G} \mathbf{S}^\top + 2\lambda \mathbf{F} (\mathbf{I}_g \odot \mathbf{F}^\top \mathbf{F})^\alpha}{\mathbf{E}_{nm} \mathbf{G} \mathbf{S}^\top + 2\lambda \mathbf{F} \mathbf{I}_g} \right)^{\frac{1}{\alpha}} \right]_+ \quad (12)$$

$$\mathbf{G}^{(t+1)} \leftarrow \mathbf{G}^{(t)} \odot \left[\left(\frac{((\mathbf{X} \odot \mathbf{F} \mathbf{S} \mathbf{G}^\top)^\top)^\alpha \mathbf{F} \mathbf{S} + 2\mu \mathbf{G} (\mathbf{I}_s \odot \mathbf{G}^\top \mathbf{G})^\alpha}{\mathbf{E}_{nm}^\top \mathbf{F} \mathbf{S} + 2\mu \mathbf{G} \mathbf{I}_s} \right)^{\frac{1}{\alpha}} \right]_+ \quad (13)$$

$$\mathbf{S}^{(t+1)} \leftarrow \mathbf{S}^{(t)} \odot \left[\left(\frac{\mathbf{F}^\top (\mathbf{X} \odot \mathbf{F} \mathbf{S} \mathbf{G}^\top)^\alpha \mathbf{G}}{\mathbf{F}^\top \mathbf{E}_{nm} \mathbf{G}} \right)^{\frac{1}{\alpha}} \right]_+, \quad (14)$$

where $[\mathbf{P}]_+ = \max(\mathbf{P}, 0)$, $\odot$ and $\oslash$ are element-wise multiplication and division, respectively. $\mathbf{E}_{nm}$ is a matrix of ones with sizes $n \times m$. Also, $\mathbf{I}_g$ and $\mathbf{I}_s$ are identity matrices with sizes $g \times g$ and $s \times s$, respectively. The parameters $\lambda \geq 0$ and $\mu \geq 0$ adjust the orthogonality of the vectors in $\mathbf{F}$ and $\mathbf{G}$.

Proof. See [Appendix D](#). □

Algorithm 1 Orthogonal Parametric Non-negative Matrix Tri-Factorization with α -divergence (OPNMTF)

Input: $\mathbf{X} \in \mathbb{R}_+^{n \times m}$ (data matrix),
 g (number of row clusters),
 s (number of column clusters),
 α (divergence parameter),
 λ (row orthogonality parameters),
 μ (column orthogonality parameters),
 N (maximum number of iterations).

Initialize: $\mathbf{F}^{(0)}$ and $\mathbf{G}^{(0)}$. Compute $\mathbf{S}^{(0)} = \mathbf{F}^{(0)\top} \mathbf{X} \mathbf{G}^{(0)}$.

- 1 While not convergent and $1 \leq t \leq N$ do:
- 2 Update $\mathbf{F}^{(t)}$ from (12),
- 3 Update $\mathbf{S}^{(t)}$ from (13),
- 4 Update $\mathbf{G}^{(t)}$ from (14),
- 5 Set $t = t + 1$.
- 6 Normalize $\mathbf{F}$, $\mathbf{S}$, and $\mathbf{G}$ with probabilistic interpretation:
- 7 $\mathbf{F} \leftarrow \mathbf{F} \text{diag}(\mathbf{S} \mathbf{D}_{\mathbf{G}} \mathbf{1}_m)$,
- 8 $\mathbf{S} \leftarrow \mathbf{D}_{\mathbf{F}} \mathbf{S} \mathbf{D}_{\mathbf{G}}^{-1}$,
- 9 $\mathbf{G} \leftarrow \mathbf{G} \text{diag}(\mathbf{1}_n^\top \mathbf{D}_{\mathbf{F}} \mathbf{S})$,
- 10 where $\mathbf{D}_{\mathbf{F}} = \text{diag}(\mathbf{1}_n^\top \mathbf{F})$ and $\mathbf{D}_{\mathbf{G}} = \text{diag}(\mathbf{1}_m^\top \mathbf{G})$.
- 11 Assign row i and column j to row and column cluster k^* and h^* if
- 12 $k^* = \arg \max_k F_{ik}, \quad h^* = \arg \max_h G_{jh}.$

Output: $\mathbf{F}$, $\mathbf{G}$, and $\mathbf{S}$.

3.1. Karush–Kuhn–Tucker conditions

The Karush–Kuhn–Tucker (KKT) conditions ([Kuhn, 2014](#)) define a first-order constraint that must be met when we solve non-linear optimization

problems. The minimization of ξ_{OPNMTF} in (8) can be defined as an inequality constrained minimization problem. It is possible to determine (12), (13), and (14) using the KKT conditions. The derivatives of ξ_{OPNMTF} over $\mathbf{F}$, $\mathbf{G}$, and $\mathbf{S}$ based on optimality and complementary slackness conditions lead us to the following equations:

$$\begin{aligned} \frac{1}{\alpha} \left(\mathbf{E}_{nm} \mathbf{G} \mathbf{S}^\top + 2\lambda \mathbf{F} \mathbf{I}_g \right) \mathbf{F}^\alpha - \frac{1}{\alpha} \left((\mathbf{X} \oslash \mathbf{F} \mathbf{S} \mathbf{G}^\top)^\alpha \mathbf{G} \mathbf{S}^\top + 2\lambda \mathbf{F} (\mathbf{I}_g \oslash \mathbf{F}^\top \mathbf{F})^\alpha \right) \mathbf{F}^\alpha &= \mathbf{0}, \\ \frac{1}{\alpha} \left(\mathbf{E}_{nm}^\top \mathbf{F} \mathbf{S} + 2\mu \mathbf{G} \mathbf{I}_s \right) \mathbf{G}^\alpha - \frac{1}{\alpha} \left(((\mathbf{X} \oslash \mathbf{F} \mathbf{S} \mathbf{G}^\top)^\top)^\alpha \mathbf{F} \mathbf{S} + 2\mu \mathbf{G} (\mathbf{I}_s \oslash \mathbf{G}^\top \mathbf{G})^\alpha \right) \mathbf{G}^\alpha &= \mathbf{0}, \\ \left(\mathbf{F}^\top \mathbf{E}_{nm} \mathbf{G} - \mathbf{F}^\top (\mathbf{X} \oslash \mathbf{F} \mathbf{S} \mathbf{G}^\top)^\alpha \mathbf{G} \right) \mathbf{S}^\alpha &= \mathbf{0}. \end{aligned}$$

Therefore, the multiplicative update rule for (8) is provided by equations (12), (13), and (14).

4. Numerical Experiments

In real text datasets, due to not having true column labels, we evaluate the quality of row clustering by one internal and three external measures. The performance of various algorithms in Table 1 is compared by three following criteria: Accuracy (Acc), Normalize Mutual Information (NMI), and Adjusted Rand Index (ARI). Also, Inter-cluster Centroids Average Similarity (ICAS) as an internal measure can be used for datasets without row and column true labels.

4.1. Evaluation Measures

Acc (Laclau and Nadif, 2016) is the basic criteria for supervised clustering and it is defined as:

$$\text{Acc} = \frac{1}{n} \max \left[\sum_{\mathcal{C}_k, \mathcal{L}_{k'}} T(\mathcal{C}_k, \mathcal{L}_{k'}) \right],$$

where $\mathcal{C}_k$, $\mathcal{L}_{k'}$, and $T(\mathcal{C}_k, \mathcal{L}_{k'}) = \mathcal{C}_k \cap \mathcal{L}_{k'}$ are sets that defines

$\mathcal{C}_k$	the prediction k -th cluster,
$\mathcal{L}_{k'}$	the true k' -th class,
$T(\mathcal{C}_k, \mathcal{L}_{k'})$	the proportion of correctly recovered objects.

NMI (Strehl and Ghosh, 2002) is calculated as follows:

$$\text{NMI} = \frac{\sum_{k,k'} \frac{n_{kk'}}{n} \log \frac{n_{kk'}}{n_k \hat{n}_{k'}}}{\sqrt{\left(\sum_k \frac{n_k}{n} \log \frac{n_k}{n} \right) \left(\sum_{k'} \frac{\hat{n}_{k'}}{n} \log \frac{\hat{n}_{k'}}{n} \right)}},$$

where n , n_k , $\hat{n}_{k'}$, and $n_{kk'}$ are integer numbers define as follows

n	# rows of data matrix $\mathbf{X}$,
n_k	# data in the cluster $\mathcal{C}_k (1 \leq k \leq g)$,
$\hat{n}_{k'}$	# data in the class $\mathcal{L}_{k'} (1 \leq k' \leq g)$,
$n_{kk'}$	# data in $\mathcal{C}_k \cap \mathcal{L}_{k'}$.

ARI (Hubert and Arabie, 1985) computes the similarity between two clustering partitions defined as follows:

$$\text{ARI} = \frac{\sum_{k,k'} \binom{n_{kk'}}{2} - \left[\sum_k \binom{n_k}{2} \sum_{k'} \binom{\hat{n}_{k'}}{2} \right] / \binom{n}{2}}{\frac{1}{2} \left[\sum_k \binom{n_k}{2} + \sum_{k'} \binom{\hat{n}_{k'}}{2} \right] - \left[\sum_k \binom{n_k}{2} \sum_{k'} \binom{\hat{n}_{k'}}{2} \right] / \binom{n}{2}}.$$

The ARI measures the similarity between estimated and true cluster labels. These three metrics have a range between 0 and 1, and a value close to 1 indicates a good clustering result.

ICAS (Del Buono and Pio, 2015) is used to measure similarity between

cluster centroids (i.e., the columns of $\mathbf{F}$) as follows:

$$\text{ICAS} = \frac{\sum_{k=1}^g \sum_{k'=k+1}^g \text{sim}(\text{centroid}_k, \text{centroid}_{k'})}{0.5 \times g \times (g-1)}, \quad (15)$$

where the centroid_k is the k -th cluster centroid, $\text{sim}(\text{centroid}_k, \text{centroid}_{k'})$ is the cosine similarity function, and g is the number of row-clusters. This measure is between 0 and 1, and the optimal value is the minimum.

4.2. Real Datasets

The proposed algorithms were compared on six text datasets, which are presented in Table 3.

Table 3: Description of datasets by topics, number of documents, and words.

Datasets	Topics	(#Documents, #Words)
CSTR (Li, 2005)	Natural Language Processing, Robotics/Vision, Systems, and Theory	(475,1000)
WebACE (Han et al., 1998)	20 different topics were obtained from the WebACE project	(2340,1000)
RCV1 (Cai and He, 2011)	It is a sub-set of a newswire stories corpus made available by Reuters containing four categories: C15, ECAT, GCAT, and MCAT.	(9625,29992)
Reviews (Karypis, 2002)	Food, Music, Movies, Radio and Restaurants.	(4069,18483)
Classic3 (Dhillon et al., 2003)	Medical, Information retrieval, and Aeronautical systems.	(3891,4303)
NG20 (Karypis, 2002)	20 different topics were obtained	(19949,43586)

4.3. Results

OPNMTF is the best algorithm for clustering performance in **CSTR** and **Classic3**. Algorithms in Table 1 are compared with OPNMTF in Table 4.

Table 4: Performance comparison of all algorithms on six real text datasets by averaging over 100 iterations.

Datasets	Metric	Algorithms						
		NBVD	ONM3F	ONMTF	ODNMTF	DNMTF	PNMTF	OPNMTF
CSTR	Acc	0.86	0.84	0.90	0.78	0.85	0.78	0.93
	NMI	0.78	0.76	0.78	0.73	0.80	0.69	0.85
	ARI	0.73	0.68	0.74	0.64	0.72	0.59	0.83
	ICAS	0.29	0.32	0.35	0.34	0.34	0.78	0.30
WebACE	Acc	0.29	0.30	0.27	0.37	0.39	0.31	0.33
	NMI	0.76	0.76	0.72	0.76	0.77	0.78	0.79
	ARI	0.43	0.44	0.40	0.44	0.46	0.48	0.49
	ICAS	0.30	0.31	0.25	0.37	0.35	0.80	0.27
RCV1	Acc	0.90	0.84	0.86	0.86	0.86	0.91	0.90
	NMI	0.80	0.73	0.76	0.74	0.74	0.83	0.81
	ARI	0.77	0.67	0.69	0.70	0.69	0.81	0.78
	ICAS	0.51	0.54	0.58	0.34	0.41	0.98	0.32
Reviews	Acc	0.52	0.51	0.69	0.56	0.58	0.54	0.65
	NMI	0.63	0.63	0.79	0.64	0.65	0.65	0.67
	ARI	0.47	0.48	0.74	0.49	0.52	0.51	0.55
	ICAS	0.58	0.58	0.55	0.64	0.53	0.97	0.56
Classic3	Acc	0.90	0.93	0.73	0.90	0.89	0.91	0.93
	NMI	0.76	0.80	0.60	0.76	0.74	0.80	0.81
	ARI	0.75	0.81	0.49	0.75	0.72	0.79	0.81
	ICAS	0.30	0.34	0.27	0.33	0.29	0.97	0.33
NG20	Acc	0.23	0.24	0.18	0.25	0.27	0.28	0.31
	NMI	0.50	0.48	0.37	0.49	0.44	0.48	0.52
	ARI	0.38	0.36	0.21	0.37	0.37	0.35	0.39
	ICAS	0.29	0.30	0.28	0.34	0.30	0.49	0.37

* A boldface number is better than the others in each row.

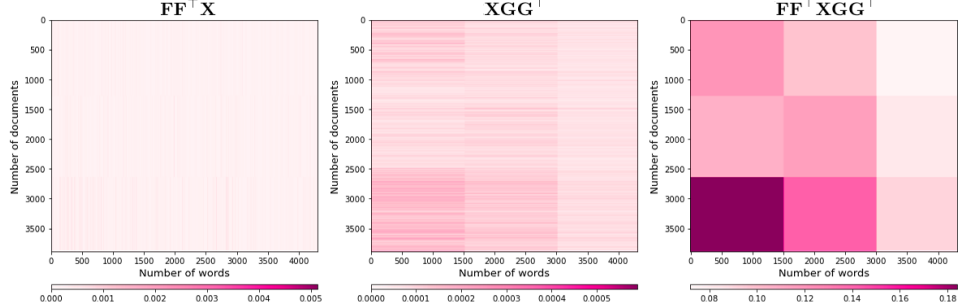


Figure 3: Result by OPNMTF with $\alpha = 1$, $\lambda = 0.03$, $\mu = 1$, $g = 3$, and $s = 3$ on **Classic3**, (Left) Clustering row data matrix ($\mathbf{FF}^T \mathbf{X}$). (Middle) Clustering column data matrix ($\mathbf{XGG}^T$). (Right) Co-clustering reorganization data matrix ($\mathbf{FF}^T \mathbf{XGG}^T$).

4.3.1. Polysemous word identification

The concept ‘polysemous’ is commonly used to define words that might have multiple meanings in different contexts. For instance, the use of the word ‘body’ with the words ‘pressure’ and ‘operation’, refers to the concept of ‘medicine’. In contrast, the use of the word ‘body’ with the words ‘wing’ and ‘aircraft’ indicates an ‘aeronautical systems’. In (Yoo and Choi, 2010), it can be seen an analysis of polysemous for **CSTR** dataset. In this subsection, we focus on **Classic3** dataset. Figure 3 displays the reorganization matrices of clustering and co-clustering by implementing the OPNMTF algorithm on **Classic3**.

Figure 1 shows the structure of the summarization matrix $\mathbf{S}$. It plays a role in identifying how Word-Cluster (WC) relates to Document-Cluster (DC). Therefore, analyzing $\mathbf{S}$ allows us to easily identify polysemous words and their corresponding contexts. Table 5 shows an example of $\mathbf{S}$ obtained by OPNMTF on **Classic3** dataset, with $g = 3$ and $s = 3$. Document-Clusters (denoted by DC1 through DC3) are connected to Word-Clusters (denoted by WC1 through WC3). Figure 4 depicts the reorganization of the

top 100 words based on Table 5.

Table 5: The summarization matrix $\mathbf{S}$ obtained by OPNMTF with $\alpha = 1$, $\lambda = 0.03$, and $\mu = 1$. DC1 to DC3 represent Document-Clusters 1 through 3, and WC1 to WC3 represent Word-Clusters 1 through 3.

	WC1	WC2	WC3
DC1: Medical	0.1173	0.0834	0.0856
DC2: Information retrieval	0.0935	0.0874	0.0886
DC3: Aeronautical systems	0.1956	0.1194	0.1287

DC1-WC1 in Figure 4a, DC2-WC2 in Figure 4e, and DC3-WC3 in Figure 4i contain words representing topics ‘medical’, ‘information retrieval’, ‘aeronautical systems’, respectively. In addition, the words in WC1, such as ‘layer’ and ‘body’ are suitable for use in DC1 as a ‘medical’ topic and DC3 as a ‘aeronautical systems’ topic.



Figure 4: Word clouds of the top 100 words for matrix $\mathbf{S}$ have been shown in Table 5 for **Classic3** dataset. The bigger word has more frequency.

4.3.2. Parameter sensitivity

In this subsection, we have two strategies for parameter sensitivities. The first plan is choosing a tuple (λ, μ) from the set $\{(10^{-9}, 10^{-9}), (10^{-9}, 10^{+9}), (10^{+9}, 10^{-9}), (10^{+9}, 10^{+9})\}$ that is defined according to the duality of very big (or small) λ with very big (or small) μ value. The second plan is choos-

ing value α from the set $\{0, 0.5, 1\}$. The results of these strategies are shown in Figures 5 and 6 for all datasets, respectively. There is a relation between orthogonality parameters and the variance of evaluation measures for each dataset. In Figure 5, the performance of evaluation measures has high variance when $(\lambda, \mu) \in \{(10^{-9}, 10^{-9}), (10^{-9}, 10^{+9})\}$, while the performance mentioned has low variance when $(\lambda, \mu) \in \{(10^{+9}, 10^{-9}), (10^{+9}, 10^{+9})\}$ for **RCV1**, **Classic3**, and **Reviews** datasets. The various values of $\alpha \in \{0.5, 1, 2\}$ are almost the same performance for all datasets in Figure 6.

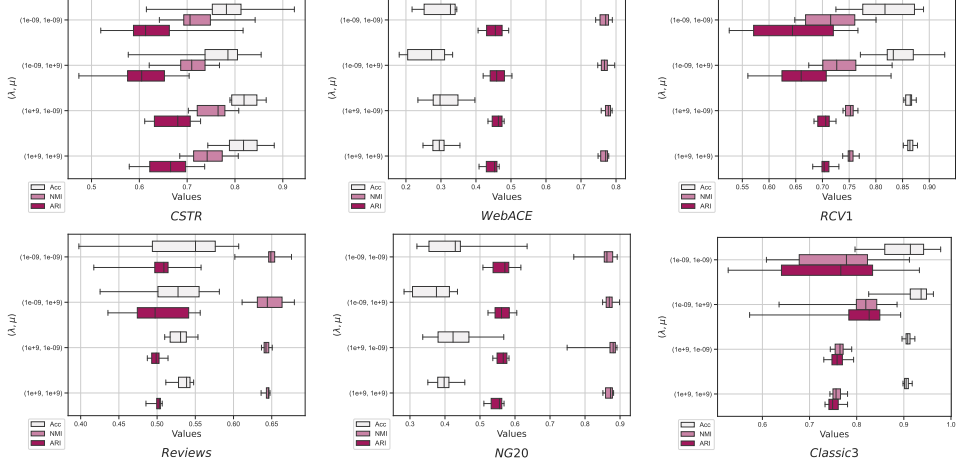


Figure 5: Box plots of measures (Acc, NMI, ARI) for six datasets are introduced in Table 3, based on 100 iterations of OPNMTF with $\alpha = 0.5$ and $(\lambda, \mu) \in \{(10^{-9}, 10^{-9}), (10^{-9}, 10^{+9}), (10^{+9}, 10^{-9}), (10^{+9}, 10^{+9})\}$.

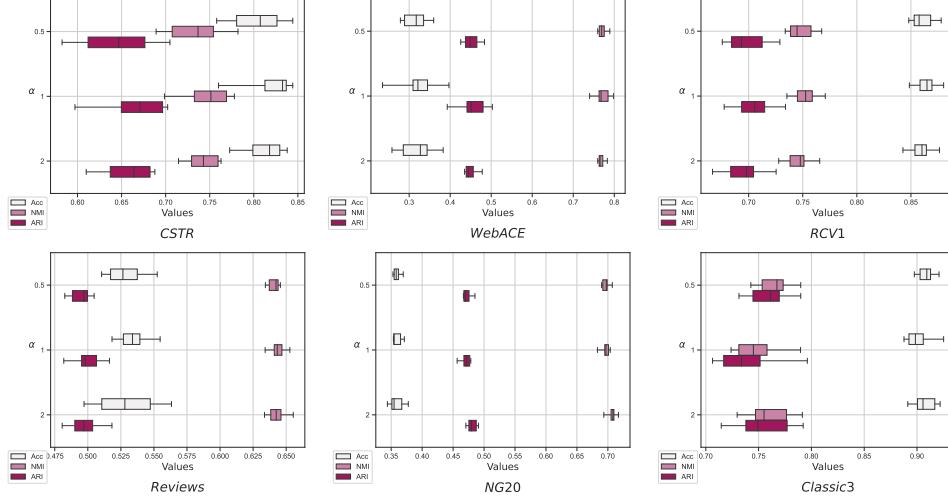


Figure 6: Box plots of measures (Acc, NMI, ARI) for six datasets are introduced in Table 3, based on 100 iterations of OPNMTF with $\lambda = 0.03$, $\mu = 1$, and $\alpha \in \{0, 0.5, 1\}$.

4.3.3. Orthogonality $\mathbf{F}$ and $\mathbf{G}$

The orthogonality measure for $\mathbf{F}$ and $\mathbf{G}$, computed by $\|\mathbf{F}^\top \mathbf{F} - \mathbf{I}_g\|$ and $\|\mathbf{G}^\top \mathbf{G} - \mathbf{I}_s\|$, as the iterations proceed. Figure 7 shows orthogonality measures based on extreme value of (λ, μ) for all datasets. The majority of datasets, such as **RCV1** and **Reviews**, have low values of orthogonality measures and low variance.

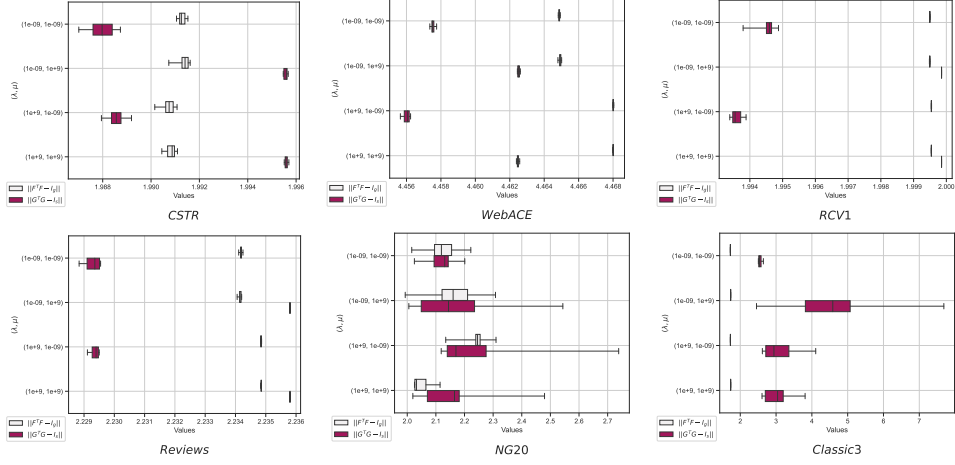


Figure 7: Box plots of orthogonality measures for $\mathbf{F}$ and $\mathbf{G}$ in six datasets are introduced in Table 3, based on 100 iterations of OPNMTF with $\alpha = 0.5$ and $(\lambda, \mu) \in \{(10^{-9}, 10^{-9}), (10^{-9}, 10^{+9}), (10^{+9}, 10^{-9}), (10^{+9}, 10^{+9})\}$.

4.3.4. Runtime algorithms

Figure 8 shows the running time list of different co-clustering algorithms in log seconds. As the number of documents and words increases, the efficacy of various co-clustering algorithms changes. For example, the proposed algorithms have a higher speed for the large datasets **RCV1** and **NG20**.

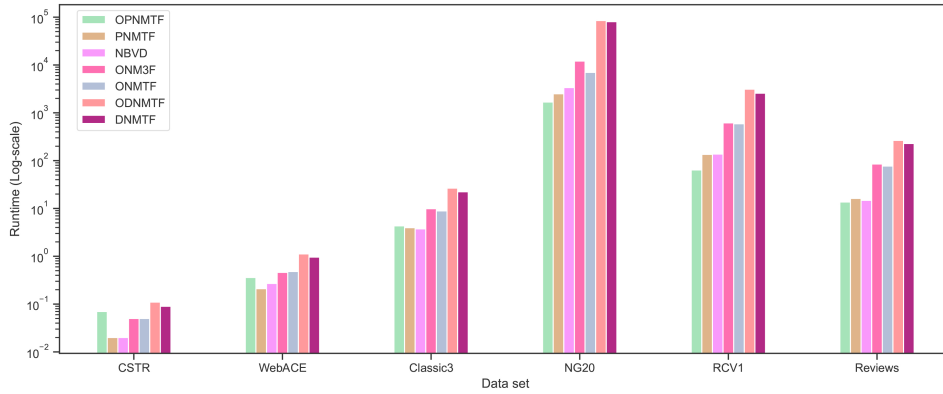


Figure 8: Runtime bar charts of different co-clustering algorithms for six datasets are reported in Table 3, based on 100 iterations for each algorithm.

5. Conclusion and future work

The OPNMTF algorithm uses optimization techniques based on α -divergence to generate an orthogonal factor matrix as part of the objective function, rather than relying on additional constraints. This approach has been tested and is effective in several real-world datasets involving text data. In this method, the values of the parameters λ and μ can be adjusted to achieve different levels of orthogonality. However, there is no set rule for determining the best values for these parameters when using the algorithm for co-clustering. In future work, it is possible to investigate more efficient algorithms for orthogonal non-negative matrix factorization problems that incorporate penalty terms.

CRedit authorship contribution statement

Saeid Hoseinipour : Conceptualization, Methodology, Investigation, Software, Visualization, Writing–original draft, Writing–review & editing. **Mina Aminghafari**: Supervision, Validation, Writing – review & editing, Funding acquisition. **Adel Mohammadpour**: Supervision, Validation, Writing–review & editing.

Data availability

The code of algorithms, all datasets, additional visualizations, and materials are available at [NMTFcoclust](#) as the repository on GitHub of the first author. Our experiments were performed on a PC (Intel(R), Core(TM) i7-10510U, 2.30 GHz), and all figures were produced in Python using the Seaborn and Matplotlib libraries.

Appendix A. Notation

Notation	Usage
$i = 1, \dots, n$	n is the number of rows
$j = 1, \dots, m$	m is the number of columns
$k = 1, \dots, g$	g is the number of row clusters
$h = 1, \dots, s$	s is the number of column clusters
$\mathbb{R}_+^{n \times m}$	Real space $n \times m$ with non-negative values
$\mathbb{R}_{\{0,1\}}^{g \times g}$	Real space $g \times g$ with values zero and one
$\mathbf{X} = (X_{ij}) \in \mathbb{R}_+^{n \times m}$	Data matrix
$\mathbf{F} = (F_{ik}) \in \mathbb{R}_+^{n \times g}$	Row cluster matrix
$\mathbf{S} = (S_{kh}) \in \mathbb{R}_+^{g \times s}$	Summary co-cluster matrix
$\mathbf{G} = (G_{jh}) \in \mathbb{R}_+^{m \times s}$	Column cluster matrix
$\mathbf{FSG}^\top = ([\mathbf{FSG}^\top]_{ij}) \in \mathbb{R}_+^{n \times m}$	Approximation matrix
$\mathbf{E}_{nm} \in \mathbb{R}_1^{n \times m}$	A matrix of ones with size $n \times m$
$\mathbf{I}_g = (I_{kk}) \in \mathbb{R}_{\{0,1\}}^{g \times g}$	Identity matrix with size $g \times g$
$\mathbf{I}_s = (I_{hh}) \in \mathbb{R}_{\{0,1\}}^{s \times s}$	Identity matrix with size $s \times s$
$\mathbf{D}_\mathbf{F} = \text{diag}(\mathbf{1}_n^\top \mathbf{F})$	Diagonal matrix normalization for $\mathbf{F}$
$\mathbf{D}_\mathbf{G} = \text{diag}(\mathbf{1}_m^\top \mathbf{G})$	Diagonal matrix normalization for $\mathbf{G}$
$\mathbf{1}_n$	Vector of ones with size $n \times 1$
$D_\alpha(\mathbf{X} \parallel \mathbf{FSG}^\top)$	α -Divergence between $\mathbf{X}$ and $\mathbf{FSG}^\top$
ξ_{OPNMTF}	Objective function for OPNMTF with α -divergence
λ	Orthogonality parameter row in OPNMTF
μ	Orthogonality parameter column in OPNMTF
α	Parameter α -divergence
t	Iteration index
N	Maximum number of iterations
$\mathcal{C}_k$	The prediction k -th cluster
$\mathcal{L}_{k'}$	The true k' -th class
$T(\mathcal{C}_k, \mathcal{L}_{k'})$	The proportion of correct recovered objects by the clustering algorithm
n_k	The number of rows in the cluster $\mathcal{C}_k (1 \leq k \leq g)$
$\hat{n}_{k'}$	The number of rows in the class $\mathcal{L}_{k'} (1 \leq k' \leq g)$
$n_{kk'}$	The number of rows in $\mathcal{C}_k \cap \mathcal{L}_{k'}$
n	The total number of documents
$\text{sim}(\text{centroid}_k, \text{centroid}_{k'})$	Cosine similarity function between k -th and k' -th cluster centroid
$\odot$	Element-wise multiplication
$\mathbf{A} \oslash \mathbf{B} = \frac{\mathbf{A}}{\mathbf{B}}$	Element-wise division matrix $\mathbf{A}$ on $\mathbf{B}$
$[\mathbf{P}]_+ = \max(\mathbf{P}, 0)$	Applying the non-negativity condition for matrix $\mathbf{P}$
$\text{Tr}(\mathbf{P})$	Trace of matrix $\mathbf{P}$

Appendix B. Proof of Lemma 2.1

Proof. Due to $A(\mathbf{F}^{(t+1)}, \mathbf{F}^{(t)})$ is an auxiliary function for $D_\alpha(\mathbf{X} \parallel \mathbf{F}^{(t+1)} \mathbf{S} \mathbf{G}^\top)$, we have

$$D_\alpha(\mathbf{X} \parallel \mathbf{F}^{(t+1)} \mathbf{S} \mathbf{G}^\top) \stackrel{(6)}{\leq} A(\mathbf{F}^{(t+1)}, \mathbf{F}^{(t)}) \stackrel{(7)}{\leq} A(\mathbf{F}^{(t)}, \mathbf{F}^{(t)}) \stackrel{(5)}{=} D_\alpha(\mathbf{X} \parallel \mathbf{F}^{(t)} \mathbf{S} \mathbf{G}^\top).$$

By repeating the updating rule in (7), we can identify the sequence of estimates that will lead to a local minimum of the cost function, such that

$$D_\alpha(\mathbf{X} \parallel \mathbf{F}^{(t_{min})} \mathbf{S} \mathbf{G}^\top) \leq \dots \leq D_\alpha(\mathbf{X} \parallel \mathbf{F}^{(t)} \mathbf{S} \mathbf{G}^\top) \leq \dots \leq D_\alpha(\mathbf{X} \parallel \mathbf{F}^{(0)} \mathbf{S} \mathbf{G}^\top).$$

□

Appendix C. Proof of Lemma 3.1

Proof. We need to show that the auxiliary function $A(\mathbf{F}, \mathbf{F}^{(t)})$ in (9) satisfies the following two conditions:

$$\begin{aligned} \text{(i)} \quad & A(\mathbf{F}, \mathbf{F}) \stackrel{(a)}{=} \xi_{\text{OPNMTF}}, \\ \text{(ii)} \quad & A(\mathbf{F}, \mathbf{F}^{(t)}) \stackrel{(9)}{=} \sum_{i,j,k,h} X_{ij} Q_{ijkh}^{\mathbf{F}^{(t)}} f \left(\frac{F_{ik} S_{kh} G_{hj}^\top}{X_{ij} Q_{ijkh}^{\mathbf{F}^{(t)}}} \right) + \lambda \sum_{i,k} Q_{ik}^{\mathbf{F}^{(t)}} f \left(\frac{F_{ki}^\top F_{ik}}{Q_{ik}^{\mathbf{F}^{(t)}}} \right) + \\ & \text{Const}_{\mathbf{G}} \stackrel{(a),(b),(c),(d)}{\geq} \sum_{i,j} X_{ij} f \left(\frac{\sum_{k,h} F_{ik} S_{kh} G_{hj}^\top}{X_{ij}} \right) + \lambda \sum_k f \left(\frac{\sum_i F_{ki}^\top F_{ik}}{I_{kk}} \right) + \\ & \text{Const}_{\mathbf{G}} \stackrel{(2),(8),(d)}{\geq} \xi_{\text{OPNMTF}}. \end{aligned}$$

These two conditions are proved by

$$\text{(a)} \quad \sum_{k,h} Q_{ijkh}^{\mathbf{F}^{(t)}} = \frac{\sum_{k,h} F_{ik}^{(t)} S_{kh} G_{hj}^\top}{\sum_{k',h'} F_{ik'}^{(t)} S_{k'h'} G_{h'j}^\top} = \frac{[\mathbf{F}^{(t)} \mathbf{S} \mathbf{G}^\top]_{ij}}{[\mathbf{F}^{(t)} \mathbf{S} \mathbf{G}^\top]_{ij}} = 1,$$

$$(b) \sum_i Q_{ik}^{\mathbf{F}^{(t)}} = \frac{\sum_i F_{ki}^{(t)\top} F_{ik}^{(t)}}{\sum_{i'} F_{ki'}^{(t)\top} F_{i'k}^{(t)}} = \frac{[\mathbf{F}^{(t)\top} \mathbf{F}^{(t)}]_{kk}}{[\mathbf{F}^{(t)\top} \mathbf{F}^{(t)}]_{kk}} = 1.$$

(c) Because of the convexity of $f(\cdot)$ in (3) and the use of Jensen's inequality,

$$\begin{aligned} & \bullet \sum_{i,j,k,h} X_{ij} Q_{ijkh}^{\mathbf{F}^{(t)}} f\left(\frac{F_{ik} S_{kh} G_{hj}^\top}{X_{ij} Q_{ijkh}^{\mathbf{F}^{(t)}}}\right) \geq \sum_{i,j} X_{ij} f\left(\frac{\sum_{k,h} F_{ik} S_{kh} G_{hj}^\top}{X_{ij}}\right), \\ & \bullet \sum_{i,k} Q_{ik}^{\mathbf{F}^{(t)}} f\left(\frac{F_{ki}^\top F_{ik}}{Q_{ik}^{\mathbf{F}^{(t)}}}\right) \geq \sum_k f\left(\frac{\sum_i F_{ki}^{(t)\top} F_{ik}^{(t)}}{I_{kk}}\right). \end{aligned}$$

$$(d) \text{Const}_{\mathbf{G}} \geq \mu D_\alpha(\mathbf{I}_s \| \mathbf{G}^\top \mathbf{G}).$$

□

Appendix D. Proof of Theorem 3.4

Proof. Note that

$$\frac{\partial f(y)}{\partial y} = \frac{1}{\alpha} (1 - y^{-\alpha}), \quad (\text{D.1})$$

the minimum of (9) is resolved by gradient equals zero using (D.1):

$$\frac{\partial A(\mathbf{F}, \mathbf{F}^{(t)})}{\partial F_{ik}} = \frac{1}{\alpha} \sum_{h,j} S_{kh} G_{hj}^\top \left(1 - \left(\frac{F_{ik} S_{kh} G_{hj}^\top}{X_{ij} Q_{ijkh}^{\mathbf{F}^{(t)}}}\right)^{-\alpha}\right) + \frac{1}{\alpha} 2\lambda \sum_k F_{ik} \left(1 - \left(\frac{F_{ki}^\top F_{ik}}{Q_{ik}^{\mathbf{F}^{(t)}}}\right)^{-\alpha}\right) = 0,$$

which gives rise to

$$\left(\frac{F_{ik}}{F_{ik}^{(t)}}\right)^\alpha = \frac{\sum_{h,j} S_{kh} G_{hj}^\top \left(\frac{X_{ij}}{\sum_{k',h'} F_{ik'}^{(t)} S_{k'h'} G_{h'j}^\top}\right)^\alpha + 2\lambda \sum_k F_{ik} \left(\frac{I_{kk}}{[\mathbf{F}^\top \mathbf{F}]_{kk}}\right)^\alpha}{\sum_{h,j} S_{kh} G_{hj}^\top + 2\lambda \sum_k F_{ik}}.$$

Therefore, we suggest the element-wise updating rule for F_{ik} is:

$$F_{ik}^{(t+1)} \leftarrow F_{ik}^{(t)} \left(\frac{\sum_{h,j} S_{kh} G_{hj}^\top \left(\frac{X_{ij}}{[\mathbf{FSG}^\top]_{ij}} \right)^\alpha + 2\lambda \sum_k F_{ik} \left(\frac{I_{kk}}{[\mathbf{F}^\top \mathbf{F}]_{kk}} \right)^\alpha}{\sum_j [\mathbf{SG}^\top]_{kj} + 2\lambda \sum_k F_{ik}} \right)^{\frac{1}{\alpha}},$$

and suggest the matrix-wise updating rule for $\mathbf{F}$

$$\mathbf{F}^{(t+1)} \leftarrow \mathbf{F}^{(t)} \odot \left[\left(\frac{(\mathbf{X} \oslash \mathbf{FSG}^\top)^\alpha \mathbf{GS}^\top + 2\lambda \mathbf{F}(\mathbf{I}_g \oslash \mathbf{F}^\top \mathbf{F})^\alpha}{\mathbf{E}_{nm} \mathbf{GS}^\top + 2\lambda \mathbf{FI}_g} \right)^{\frac{1}{\alpha}} \right]_+,$$

that is identical to (12).

similarity, the updating rule (13) and (14) are determined by solving $\frac{\partial A(\mathbf{G}, \mathbf{G}^{(t)})}{\partial G_{jh}}$ and $\frac{\partial A(\mathbf{S}, \mathbf{S}^{(t)})}{\partial S_{kh}}$, where $A(\mathbf{G}, \mathbf{G}^{(t)})$ and $A(\mathbf{S}, \mathbf{S}^{(t)})$ are given in (10) and (11). $\square$

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